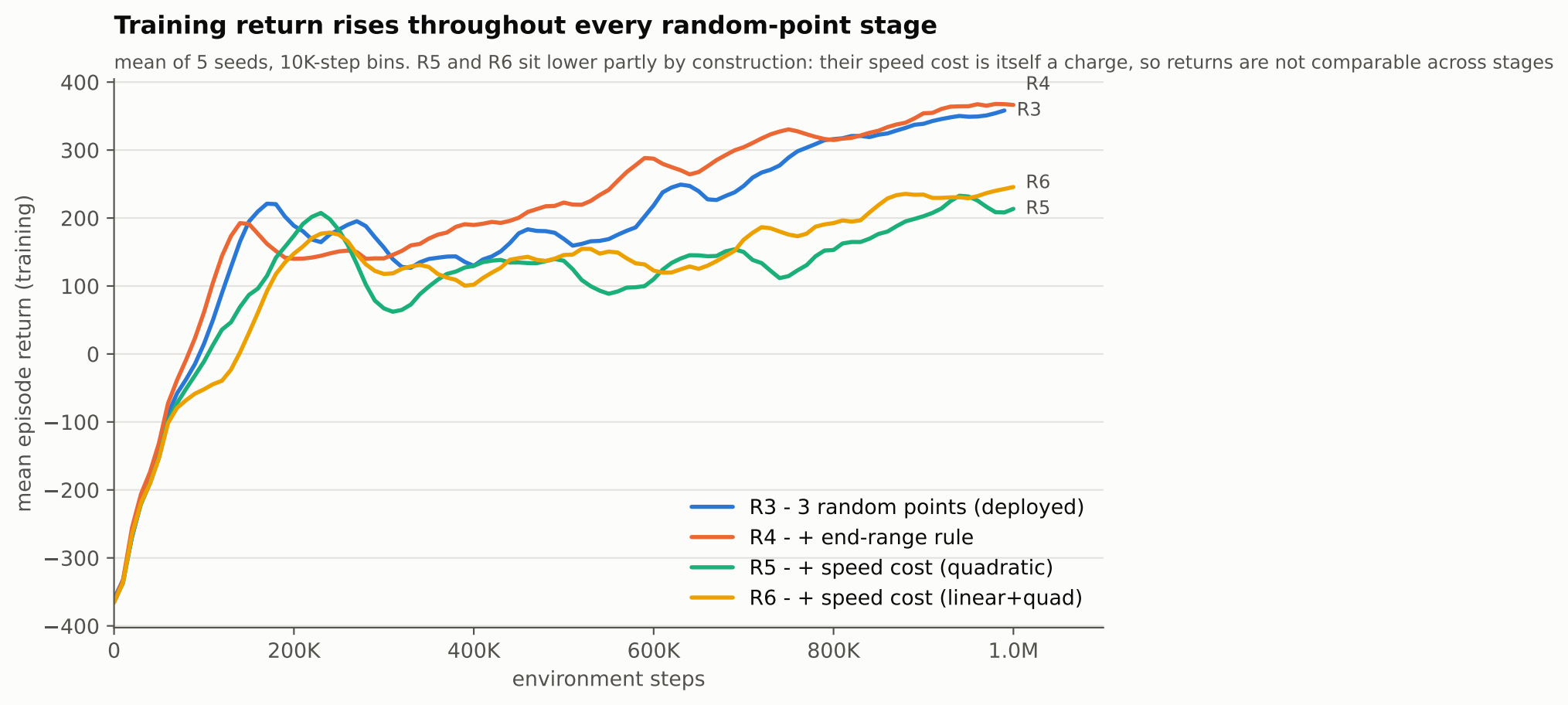
Training return rises throughout every random-point stage
mean of 5 seeds, 10K-step bins. R5 and R6 sit lower partly by construction: their speed cost is itself a charge, so returns are not comparable across stages
mean episode return (training)
environment steps
400
300
200
100
0
−100
−200
−300
−400
0
200K
400K
600K
800K
1.0M
R4
R3
R6
R5
R3 - 3 random points (deployed)
R4 - + end-range rule
R5 - + speed cost (quadratic)
R6 - + speed cost (linear+quad)